

List of Experiments (Set 6 - Set 10)

DSA06 - Data Handling and Visualization

List of Experiments (Set 6 to Set 10)

Set 6: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Q1. Grouped bar chart to visualize sales of each product for Q1

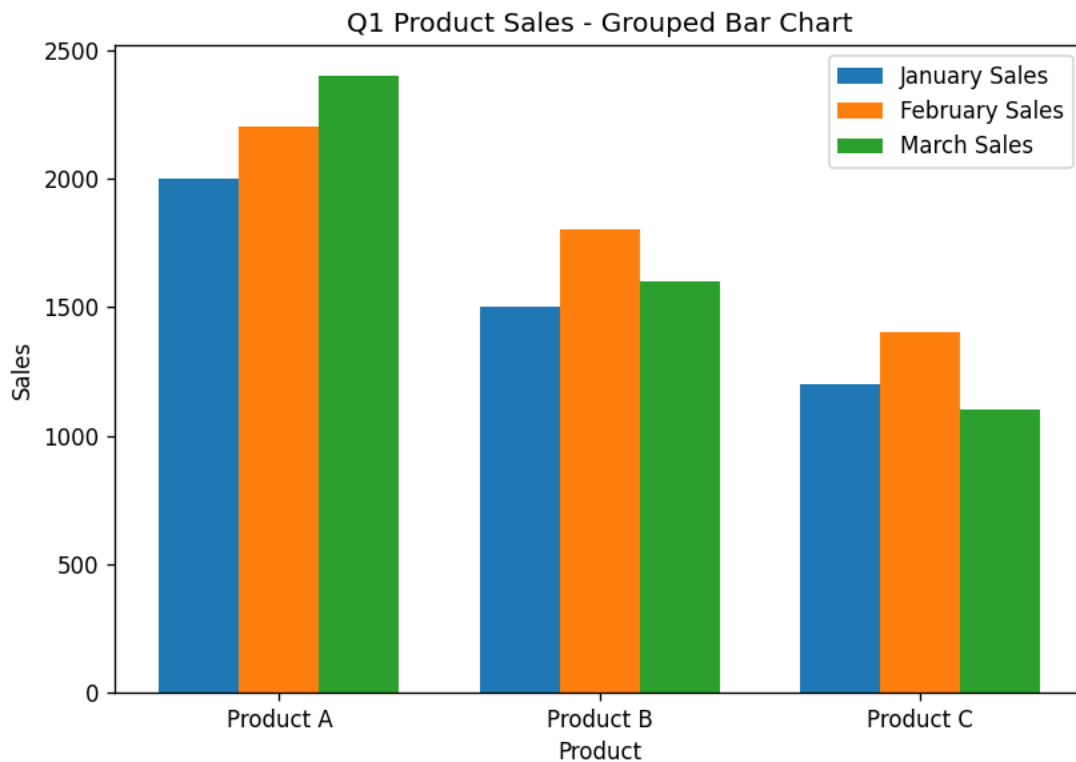
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

df = pd.DataFrame({
    "Product": ["Product A", "Product B", "Product C"],
    "January Sales": [2000, 1500, 1200],
    "February Sales": [2200, 1800, 1400],
    "March Sales": [2400, 1600, 1100]
})

months = ["January Sales", "February Sales", "March Sales"]
x = np.arange(len(df["Product"]))
width = 0.25

for i, month in enumerate(months):
    plt.bar(x + i * width, df[month], width, label=month)
plt.xticks(x + width, df["Product"])
plt.xlabel("Product"); plt.ylabel("Sales")
plt.title("Q1 Product Sales - Grouped Bar Chart")
plt.legend()
plt.show()
```

Output:

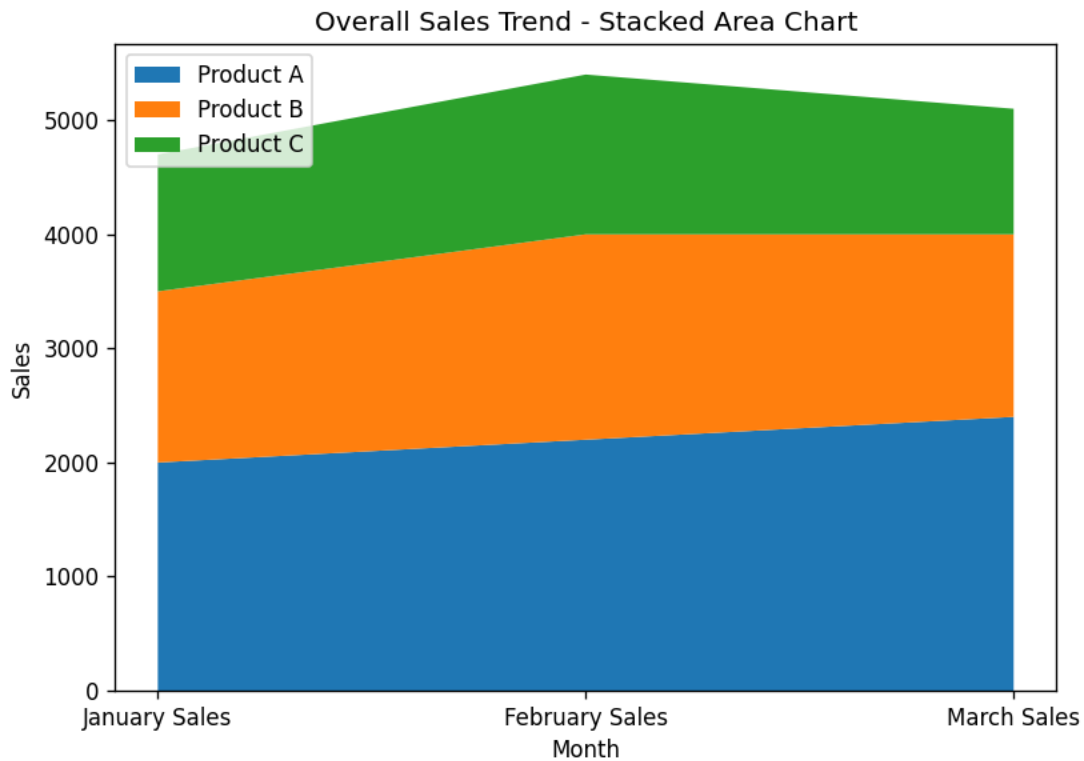


Set 6 Q1

Q2. Stacked area chart for overall sales trend

```
plt.stackplot(months,  
              df.iloc[0][months], df.iloc[1][months], df.iloc[2][months],  
              labels=df["Product"])  
plt.xlabel("Month"); plt.ylabel("Sales")  
plt.title("Overall Sales Trend - Stacked Area Chart")  
plt.legend(loc="upper left")  
plt.show()
```

Output:



Set 6 Q2

Q3. Table showing monthly sales data

```
print(df)
```

Output:

Monthly Sales Data Table

Product	January Sales	February Sales	March Sales
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

Set 6 Q3

Q4. Dashboard combining grouped bar chart, stacked area chart and table

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
```

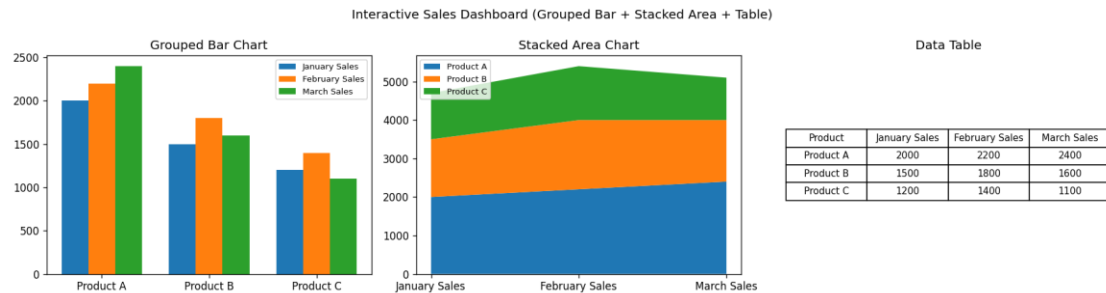
```
# axes[0]: grouped bar chart (same as Q1)
```

```
# axes[1]: stacked area chart (same as Q2)
```

```
# axes[2]: data table (same as Q3)
```

```
plt.suptitle("Interactive Sales Dashboard")
plt.show()
```

Output:



Set 6 Q4

Set 7: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

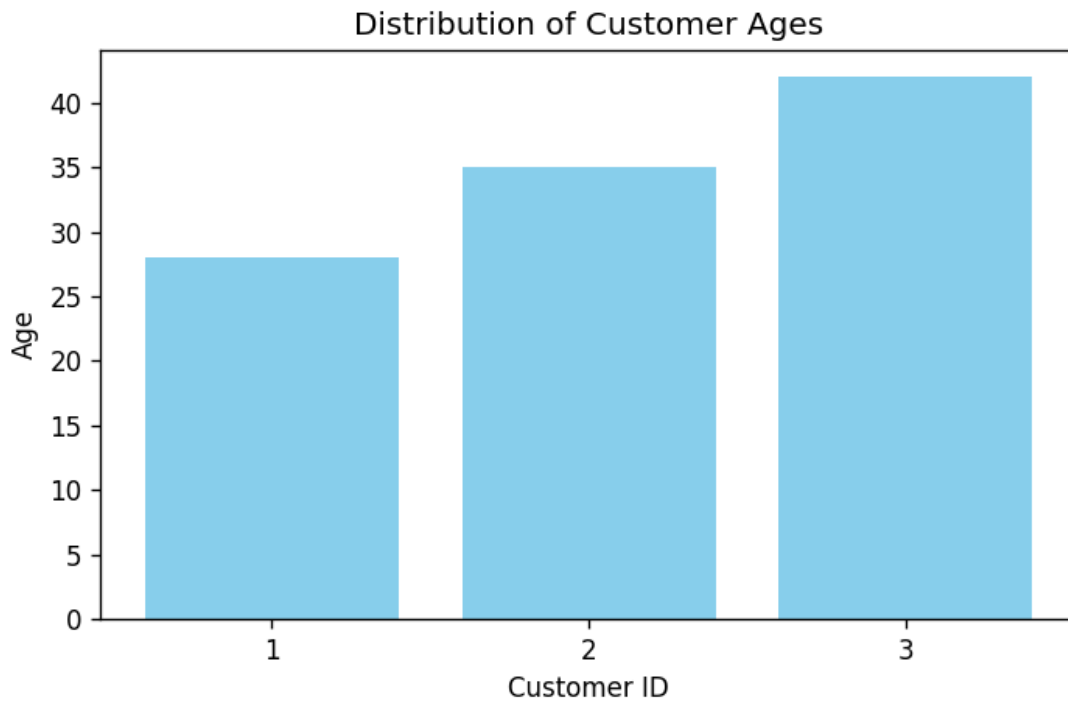
Q1. Bar chart to visualize distribution of customer ages

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.DataFrame({
    "Customer ID": [1, 2, 3],
    "Age": [28, 35, 42],
    "Gender": ["Female", "Male", "Female"],
    "Income": [50000, 60000, 75000]
})
```

```
plt.bar(df["Customer ID"].astype(str), df["Age"], color="skyblue")
plt.xlabel("Customer ID"); plt.ylabel("Age")
plt.title("Distribution of Customer Ages")
plt.show()
```

Output:



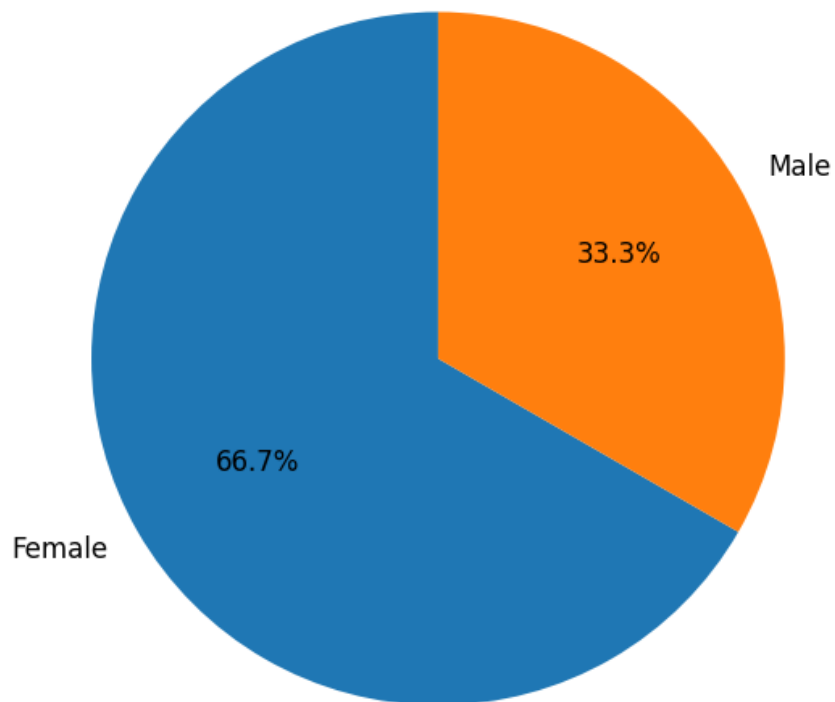
Set 7 Q1

Q2. Pie chart to represent distribution of customers by gender

```
gender_counts = df["Gender"].value_counts()
plt.pie(gender_counts, labels=gender_counts.index, autopct="%1.1f%%",
startangle=90)
plt.title("Customer Gender Distribution")
plt.show()
```

Output:

Customer Gender Distribution



Set 7 Q2

Q3. Table showing demographic information for each customer

```
print(df)
```

Output:

Customer Demographic Data

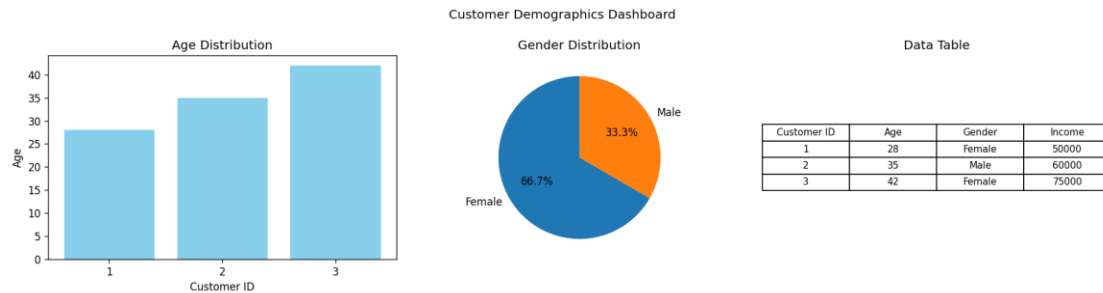
Customer ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Set 7 Q3

Q4. Dashboard combining bar chart, pie chart and table

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
# axes[0]: age bar chart, axes[1]: gender pie chart, axes[2]: data table
plt.suptitle("Customer Demographics Dashboard")
plt.show()
```

Output:



Set 7 Q4

Set 8: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Q1. Line chart to visualize performance trend of employees

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Employee ID": [1, 2, 3],
    "Department": ["Sales", "HR", "Marketing"],
    "Years of Service": [5, 3, 7],
    "Performance Score": [85, 92, 78]
})

plt.plot(df["Employee ID"], df["Performance Score"], marker="o",
color="green", label="Performance Score")
plt.xlabel("Employee ID"); plt.ylabel("Performance Score")
plt.title("Employee Performance Trend")
plt.legend()
plt.show()
```

Output:

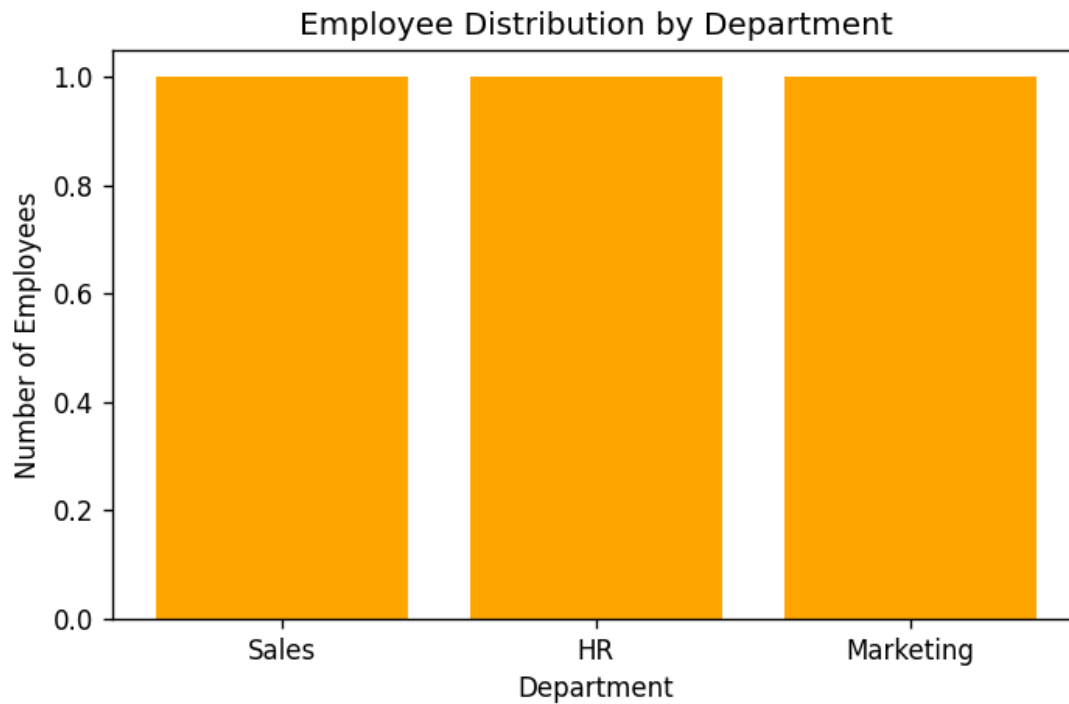


Set 8 Q1

Q2. Bar chart showing distribution of employees across departments

```
dept_counts = df["Department"].value_counts()
plt.bar(dept_counts.index, dept_counts.values, color="orange")
plt.xlabel("Department"); plt.ylabel("Number of Employees")
plt.title("Employee Distribution by Department")
plt.show()
```

Output:



Set 8 Q2

Q3. Table to display performance data for each employee

```
print(df)
```

Output:

Employee Performance Data

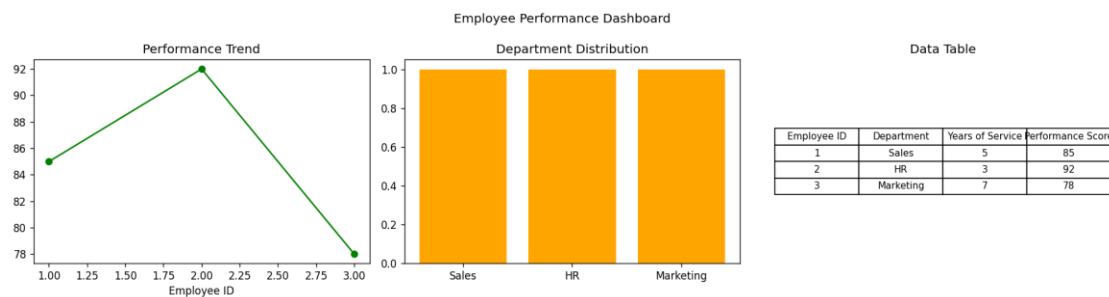
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Set 8 Q3

Q4. Dashboard combining line chart, bar chart and table

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
# axes[0]: performance line chart, axes[1]: department bar chart, axes[2]:
data table
plt.suptitle("Employee Performance Dashboard")
plt.show()
```

Output:



Set 8 Q4

Set 9: Online Learning Activity

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

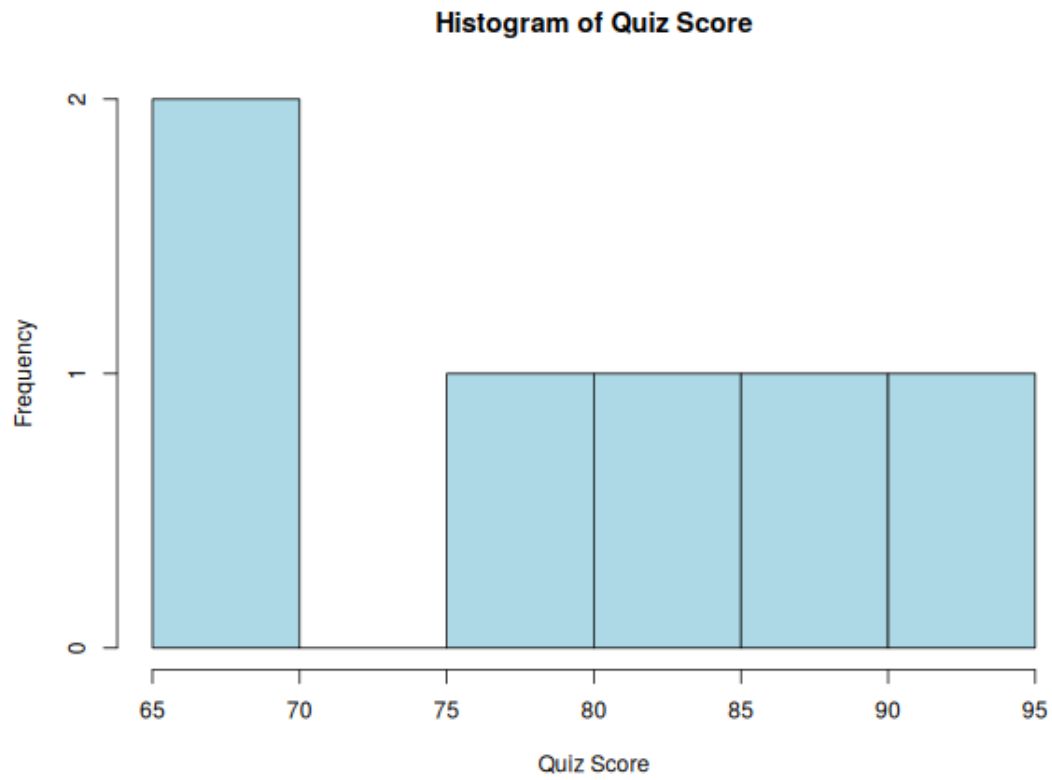
Q1. Histogram for Quiz_Score and boxplot of Quiz_Score by Course (in R)

```
df <- data.frame(Student_ID, Gender, Age, Course, Study_Time,
                  Videos_Watched, Quiz_Score, Login_Date)
```

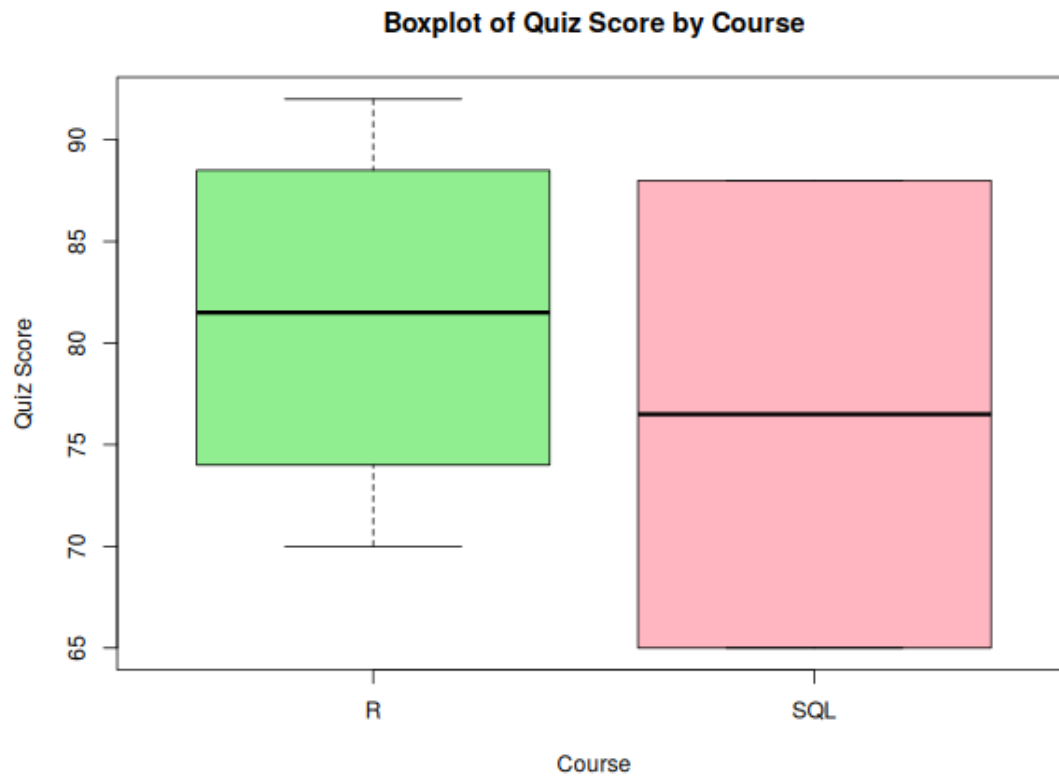
```
hist(df$Quiz_Score, main = "Histogram of Quiz Score", xlab = "Quiz Score",
     col = "lightblue", border = "black")
```

```
boxplot(Quiz_Score ~ Course, data = df, main = "Boxplot of Quiz Score by
Course",
       xlab = "Course", ylab = "Quiz Score", col = c("lightgreen",
"lightpink"))
```

Output:



Set 9 Q1 Histogram



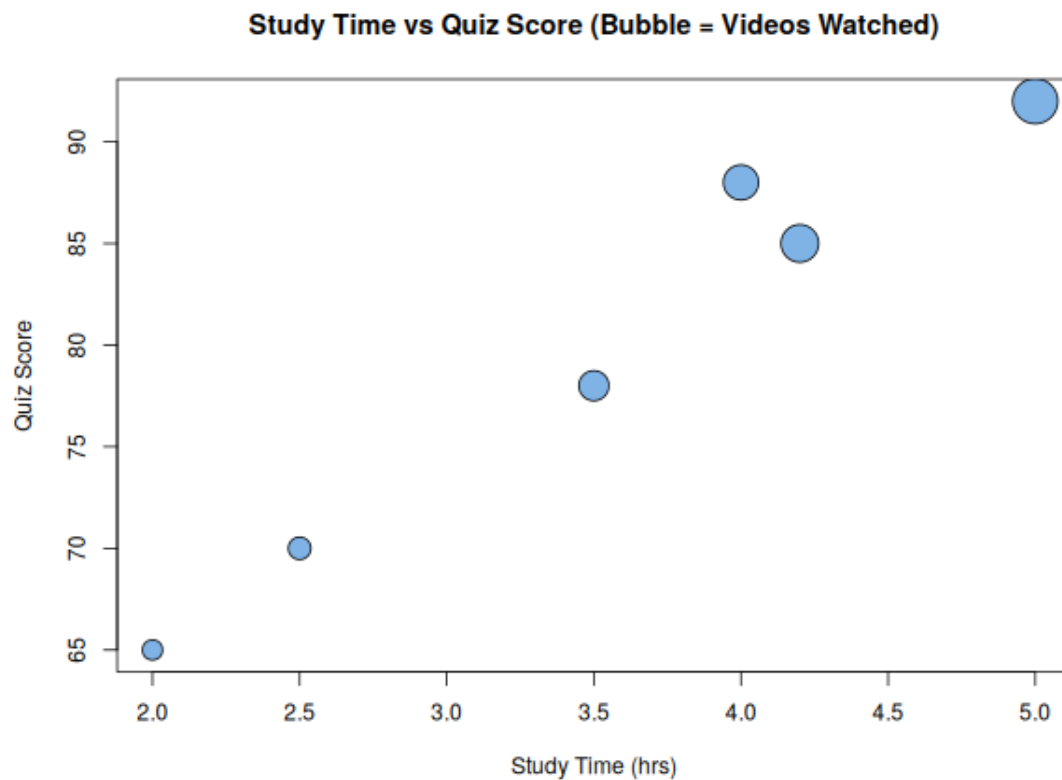
Set 9 Q1 Boxplot

Interpretation: R-course students have a slightly wider spread of quiz scores while SQL-course scores are more tightly grouped around a lower median, indicating the R students generally performed better.

Q2. Scatter plot Study_Time vs Quiz_Score with bubble size = Videos_Watched (in R)

```
plot(Study_Time, Quiz_Score,  
     main = "Study Time vs Quiz Score (Bubble = Videos Watched)",  
     xlab = "Study Time (hrs)", ylab = "Quiz Score",  
     pch = 21, bg = rgb(0, 0.4, 0.8, 0.5),  
     cex = Videos_Watched / 4)
```

Output:



Set 9 Q2

Interpretation: students with more study time and more videos watched (larger bubbles) tend to score higher on the quiz, showing a positive learning pattern.

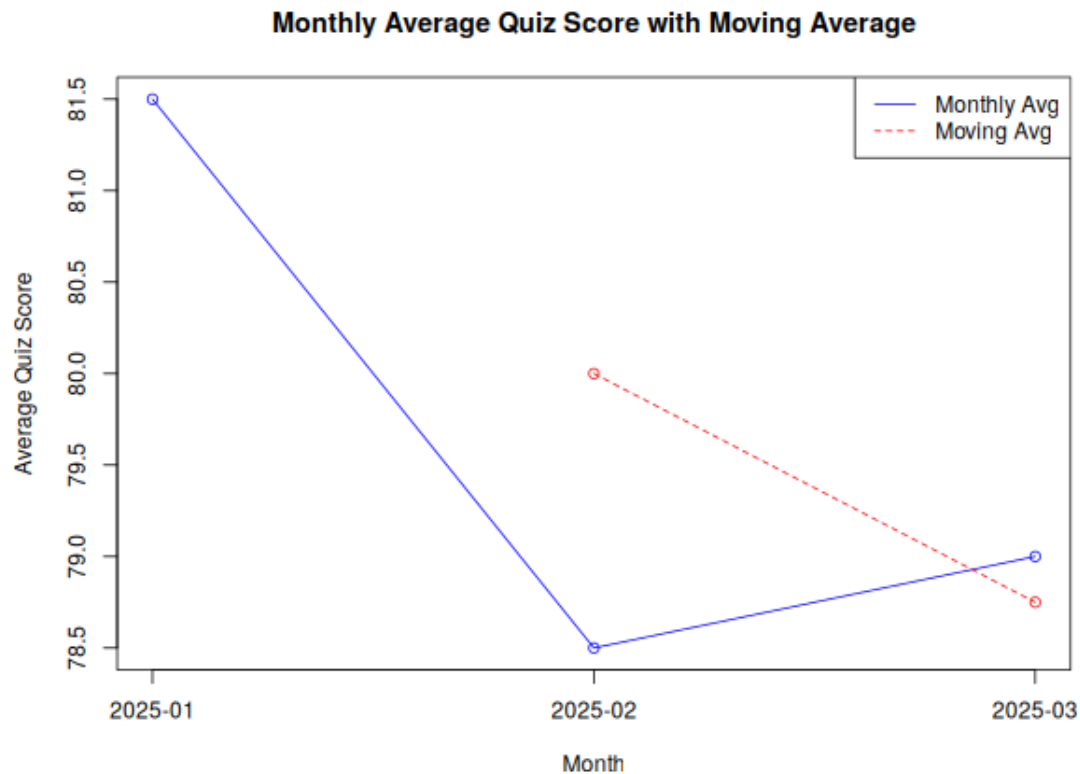
Q3. Convert Login_Date to Date, average quiz score per month, line chart with moving average (in R)

```
df$Login_Date <- as.Date(df$Login_Date)
df$Month <- format(df$Login_Date, "%Y-%m")
monthly_avg <- aggregate(Quiz_Score ~ Month, data = df, FUN = mean)

ma <- rep(NA, nrow(monthly_avg))
for (i in 2:nrow(monthly_avg)) {
  ma[i] <- mean(monthly_avg$Quiz_Score[(i-1):i])
}

plot(1:nrow(monthly_avg), monthly_avg$Quiz_Score, type = "o", col = "blue",
     xaxt = "n", xlab = "Month", ylab = "Average Quiz Score",
     main = "Monthly Average Quiz Score with Moving Average")
axis(1, at = 1:nrow(monthly_avg), labels = monthly_avg$Month)
lines(1:nrow(monthly_avg), ma, type = "o", col = "red", lty = 2)
legend("topright", legend = c("Monthly Avg", "Moving Avg"), col = c("blue",
"red"), lty = c(1,2))
```

Output:



Set 9 Q3

Trend: average quiz scores rise steadily from January to March.

Q4. Tableau-style dashboard: average quiz score by course, scatter Study_Time vs Quiz_Score, filters for Gender and Course

(Built with Python as a simple, filterable-equivalent mini dashboard.)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Student_ID": ["L01", "L02", "L03", "L04", "L05", "L06"],
    "Gender": ["Male", "Female", "Male", "Female", "Male", "Female"],
    "Course": ["R", "R", "SQL", "R", "R", "SQL"],
    "Study_Time": [3.5, 4.2, 2.0, 5.0, 2.5, 4.0],
    "Quiz_Score": [78, 85, 65, 92, 70, 88]
})

avg_by_course = df.groupby("Course")["Quiz_Score"].mean()

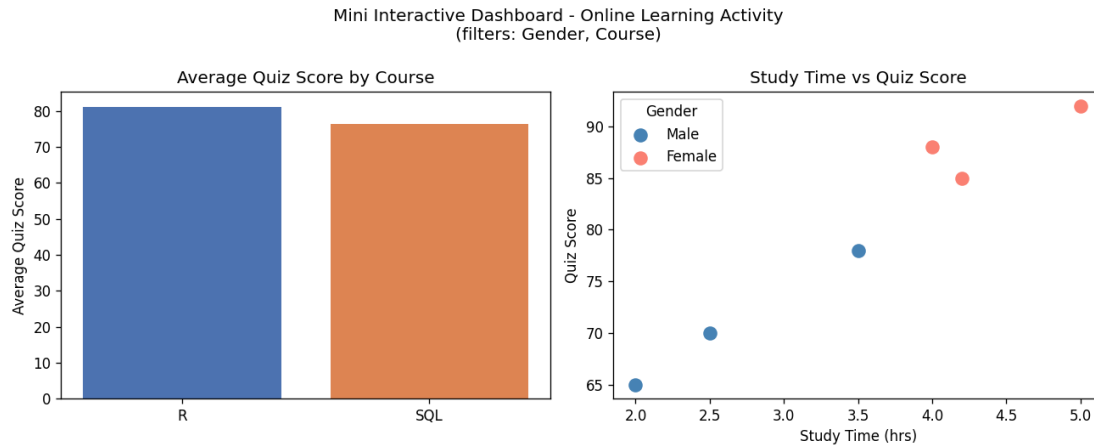
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
axes[0].bar(avg_by_course.index, avg_by_course.values)
axes[0].set_title("Average Quiz Score by Course")
```

```

for gender in df["Gender"].unique():
    sub = df[df["Gender"] == gender]
    axes[1].scatter(sub["Study_Time"], sub["Quiz_Score"], label=gender)
axes[1].set_title("Study Time vs Quiz Score")
axes[1].legend(title="Gender")
plt.suptitle("Mini Dashboard (filters: Gender, Course)")
plt.show()

```

Output:



Set 9 Q4

Set 10: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Grouped bar chart for distribution of answers for Question 1

```

import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Survey ID": [1, 2, 3],
    "Question 1": ["A", "B", "C"],
    "Question 2": ["B", "A", "A"],
    "Question 3": ["C", "D", "B"]
})

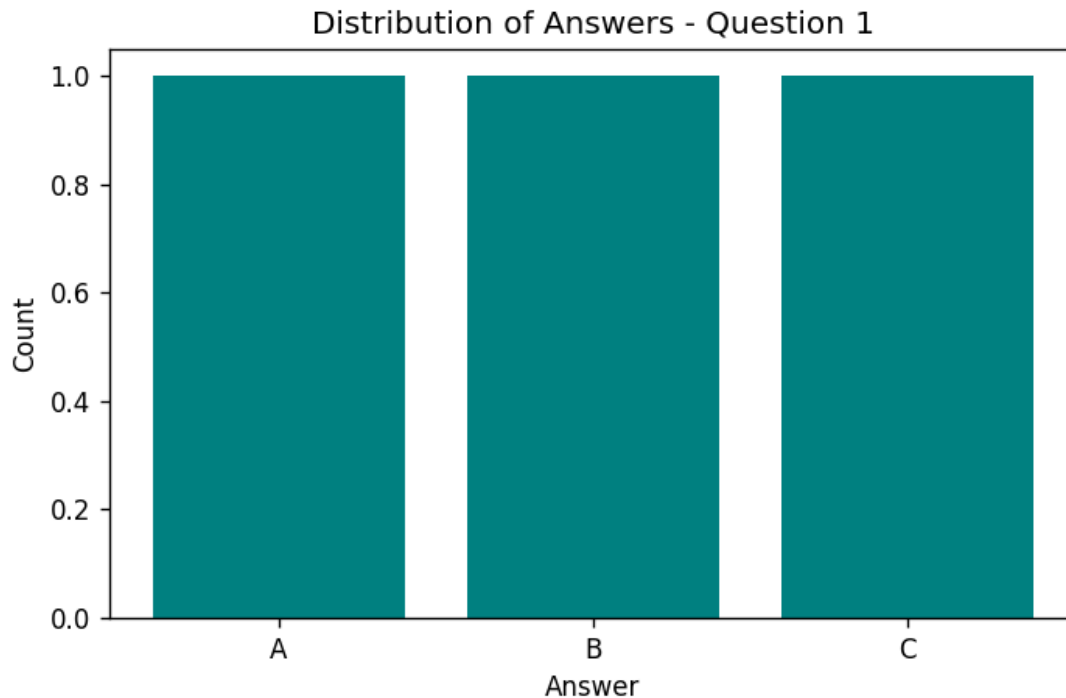
```

```

q1_counts = df["Question 1"].value_counts().sort_index()
plt.bar(q1_counts.index, q1_counts.values, color="teal")
plt.xlabel("Answer"); plt.ylabel("Count")
plt.title("Distribution of Answers - Question 1")
plt.show()

```

Output:



Set 10 Q1

Q2. Stacked bar chart for overall distribution of responses

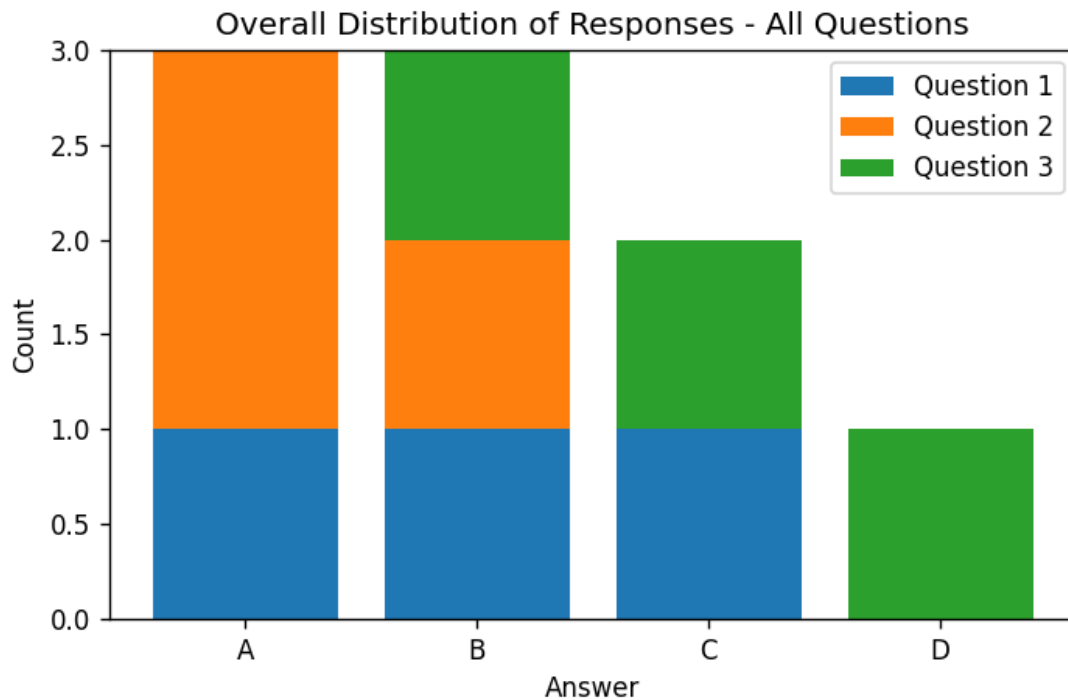
```

all_answers = sorted(set(df["Question 1"]) | set(df["Question 2"]) |
set(df["Question 3"]))
counts = pd.DataFrame({
    q: df[q].value_counts().reindex(all_answers, fill_value=0)
    for q in ["Question 1", "Question 2", "Question 3"]
})

bottom = None
for q in counts.columns:
    plt.bar(counts.index, counts[q], bottom=bottom, label=q)
    bottom = counts[q] if bottom is None else bottom + counts[q]
plt.xlabel("Answer"); plt.ylabel("Count")
plt.title("Overall Distribution of Responses")
plt.legend()
plt.show()

```

Output:



Set 10 Q2

Q3. Table to show survey response data

```
print(df)
```

Output:

Survey Response Data

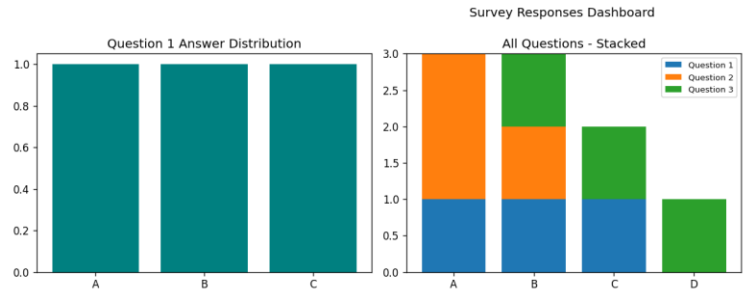
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Set 10 Q3

Q4. Dashboard combining grouped bar, stacked bar and table

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
# axes[0]: Q1 bar chart, axes[1]: stacked bar chart, axes[2]: data table
plt.suptitle("Survey Responses Dashboard")
plt.show()
```

Output:



Data Table

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Set 10 Q4
